

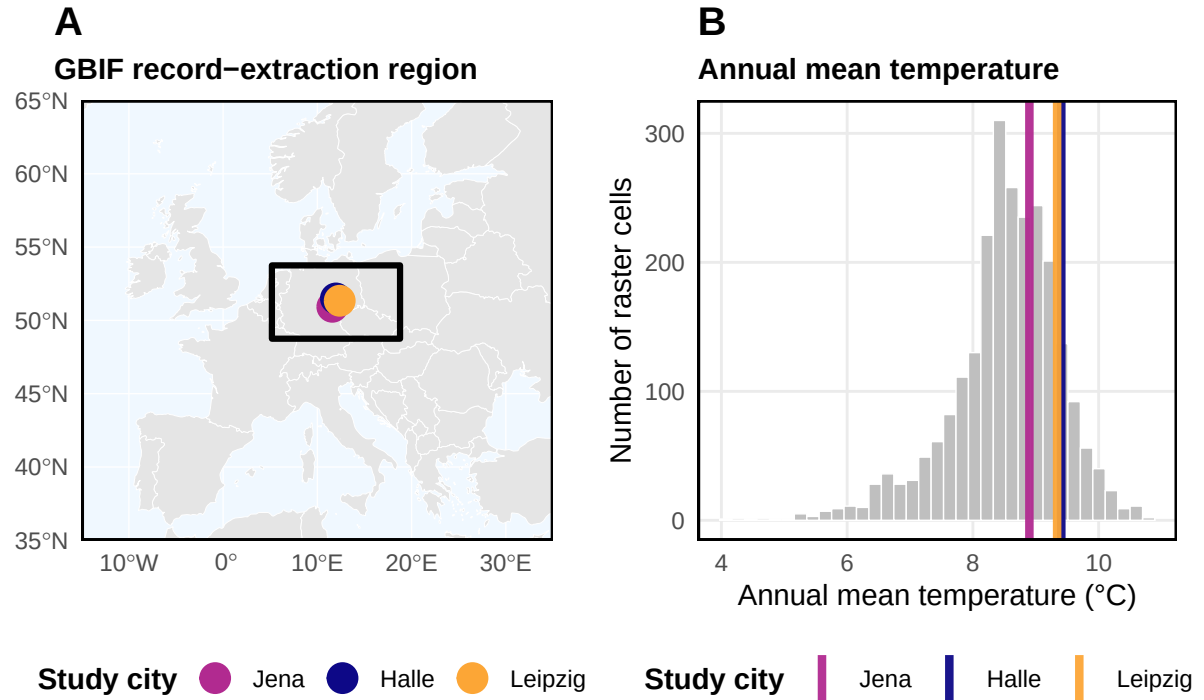


Figure S4. Spatial and climatic characterization of the GBIF record-extraction region used to estimate pollinator phenology. (A) Geographic extent (black rectangle) centred on Jena, Halle, and Leipzig. (B) Distribution of long-term annual mean temperature (WorldClim v2.1 climatological normals) within the extraction region. Colored vertical lines denote the temperatures of the three study cities, supporting the use of the selected region as a compromise between maximizing occurrence records and limiting climatic heterogeneity.